

ABCD ADHD menarche analysis: Output for cross-sectional models

Model 2 no interaction output

```
      Estimate Est.Error  Q2.5 Q97.5
R2      0.838    0.00245 0.834 0.843
```

```
Family: poisson
Links: mu = log
Formula: internalising ~ menarche_status_p + adhd_status_life + age_sc + ethnrace + inr_sc +
  Data: analytic_df (Number of observations: 19375)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~family_id (Number of levels: 4620)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.762	0.017	0.727	0.793	1.001	1026	1568

~obs_id (Number of levels: 19375)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.446	0.006	0.434	0.459	1.000	1648	2606

~participant_id (Number of levels: 5291)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.449	0.021	0.408	0.491	1.003	578	1220

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.119	0.026	0.076	0.178	1.001	3206	2916

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.071	0.034	1.006	1.136	1.001	3194	2992
menarche_status_pY	0.080	0.017	0.046	0.113	1.001	5410	2650
adhd_status_lifeY	0.624	0.033	0.558	0.689	1.001	4578	3110
age_sc	0.053	0.015	0.023	0.083	1.000	4856	3283
ethnraceblack_nh	-0.401	0.045	-0.487	-0.312	1.001	4262	2993
ethnracehispanic	-0.069	0.042	-0.152	0.012	1.000	3989	2811
ethnraceother	-0.019	0.046	-0.109	0.071	1.000	4353	3285
inr_sc	-0.048	0.020	-0.088	-0.009	1.001	5859	3572

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 2 no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable      Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>          <dbl>         <dbl>         <dbl>
1 Menarche (yes vs no)      8.3          4.67          12.0
2 ADHD (yes vs no)        86.6          74.7          99.1
```

Model 2B no interaction output

```
Estimate Est.Error Q2.5 Q97.5
R2      0.839      0.00251 0.834 0.843
```

```
Family: poisson
Links: mu = log
Formula: internalising ~ breast_development_p_sc + adhd_status_life + age_sc + ethnrace + in
Data: analytic_df (Number of observations: 19358)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family_id (Number of levels: 4613)

```
Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept) 0.759      0.018      0.725      0.793 1.000      1775      3091
```

~obs_id (Number of levels: 19358)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.446	0.006	0.434	0.459	1.000	2892	4418

~participant_id (Number of levels: 5283)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.450	0.022	0.408	0.494	1.001	934	1979

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.118	0.025	0.077	0.173	1.000	6409	6158

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	1.095	0.034	1.028	1.163	1.000	7145
breast_development_p_sc	0.077	0.018	0.041	0.113	1.000	10201
adhd_status_lifeY	0.622	0.033	0.556	0.686	1.000	7434
age_sc	0.058	0.015	0.028	0.088	1.000	10342
ethnraceblack_nh	-0.392	0.045	-0.480	-0.304	1.000	8500
ethnracehispanic	-0.061	0.041	-0.142	0.020	1.001	8823
ethnraceother	-0.019	0.046	-0.110	0.072	1.000	9127
inr_sc	-0.042	0.020	-0.083	-0.003	1.000	12251

	Tail_ESS
Intercept	5760
breast_development_p_sc	6022
adhd_status_lifeY	5665
age_sc	6852
ethnraceblack_nh	6190
ethnracehispanic	6207
ethnraceother	6189
inr_sc	6624

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 2B no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

A tibble: 2 x 4

Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development (per 1-unit)	4.87	2.52	7.18

2 ADHD (yes vs no)	86.2	74.3	98.7
--------------------	------	------	------

Model 3 no interaction output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.859	0.0025	0.854	0.864

Family: poisson
 Links: mu = log
 Formula: externalising ~ menarche_status_p + adhd_status_life + age_sc + ethnrace + inr_sc +
 Data: analytic_df (Number of observations: 19375)
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 4620)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.864	0.031	0.801	0.922	1.004	466	825

~obs_id (Number of levels: 19375)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.497	0.008	0.481	0.512	1.002	1654	2544

~participant_id (Number of levels: 5291)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.769	0.031	0.709	0.832	1.005	392	632

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.153	0.033	0.097	0.230	1.000	2301	2720

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.121	0.046	0.029	0.208	1.000	2090	2682
menarche_status_pY	0.092	0.022	0.050	0.136	1.000	3152	2800
adhd_status_lifeY	1.325	0.045	1.239	1.412	1.000	3196	2864
age_sc	-0.132	0.020	-0.172	-0.094	1.002	3679	3263
ethnraceblack_nh	-0.005	0.058	-0.123	0.107	1.001	3153	2889
ethnracehispanic	-0.011	0.055	-0.121	0.095	1.000	3128	2664
ethnraceother	-0.010	0.060	-0.132	0.109	1.000	3406	3002
inr_sc	-0.185	0.027	-0.236	-0.132	1.000	4237	3041

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 3 no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable      Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>          <dbl>         <dbl>         <dbl>
1 Menarche (yes vs no)    9.67           5.14          14.6
2 ADHD (yes vs no)      276.           245.          311.
```

Model 3B no interaction output

Estimate Est.Error Q2.5 Q97.5
R2 0.859 0.00255 0.854 0.864

Family: poisson
Links: mu = log
Formula: externalising ~ breast_development_p_sc + adhd_status_life + age_sc + ethnrace + inr_sc
Data: analytic_df (Number of observations: 19358)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 4613)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.859	0.030	0.799	0.916	1.001	1126	2143

~obs_id (Number of levels: 19358)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.494	0.008	0.478	0.510	1.000	3281	5306

~participant_id (Number of levels: 5283)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.769	0.030	0.709	0.828	1.002	988	2061

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.154	0.035	0.097	0.232	1.001	5494	5542

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	0.156	0.045	0.069	0.245	1.001	4706
breast_development_p_sc	0.129	0.022	0.085	0.173	1.000	7417
adhd_status_lifeY	1.321	0.045	1.231	1.408	1.001	6455
age_sc	-0.149	0.019	-0.186	-0.112	1.000	7591
ethnraceblack_nh	-0.013	0.058	-0.129	0.101	1.000	6814
ethnracehispanic	-0.005	0.055	-0.112	0.101	1.000	6693
ethnraceother	-0.012	0.061	-0.129	0.109	1.000	7081
inr_sc	-0.175	0.027	-0.228	-0.122	1.001	9093
	Tail_ESS					
Intercept		5408				
breast_development_p_sc		6157				

adhd_status_lifeY	5601
age_sc	6326
ethnraceblack_nh	5959
ethnracehispanic	6115
ethnraceother	5793
inr_sc	6404

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 3B Effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 2 x 4
  Variable      Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>          <dbl>         <dbl>         <dbl>
1 Breast development (per 1-unit)  8.26          5.38          11.2
2 ADHD (yes vs no)                275.          243.          309.
```

Model 4B no interaction output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.841	0.00242	0.836	0.845

```
Family: poisson
Links: mu = log
Formula: internalising ~ breast_development_p_sc + adhd_traits_sc + age_sc + ethnrace + inr_
Data: analytic_df (Number of observations: 19364)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

```
~family_id (Number of levels: 4619)
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.649	0.018	0.614	0.683	1.013	421	1290

```
~obs_id (Number of levels: 19364)
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.366	0.006	0.353	0.378	1.000	2796	4515

```

~participant_id (Number of levels: 5289)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.461    0.020   0.422   0.500 1.018    295    956

```

```

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.093    0.022   0.056   0.141 1.001   2334   3055

```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	1.221	0.028	1.165	1.276	1.001	2375
breast_development_p_sc	0.059	0.016	0.027	0.091	1.000	3752
adhd_traits_sc	0.827	0.014	0.799	0.855	1.000	3449
age_sc	0.070	0.014	0.042	0.097	1.000	3877
ethnraceblack_nh	-0.431	0.041	-0.510	-0.349	1.000	2943
ethnracehispanic	-0.075	0.036	-0.147	-0.003	1.002	2543
ethnraceother	-0.046	0.041	-0.128	0.035	1.000	2532
inr_sc	-0.019	0.019	-0.056	0.018	1.001	4021

	Tail_ESS
Intercept	3851
breast_development_p_sc	4837
adhd_traits_sc	4778
age_sc	5126
ethnraceblack_nh	4012
ethnracehispanic	4500
ethnraceother	3954
inr_sc	4829

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 4B no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```

# A tibble: 2 x 4
  Variable      Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>      <dbl>      <dbl>      <dbl>
1 Breast development (per 1-unit)    3.69         1.66         5.75
2 ADHD traits (per 1-unit)         15.8        15.2        16.4

```


Model 5B no interaction output

```

      Estimate Est.Error  Q2.5 Q97.5
R2      0.863      0.00234 0.858 0.867

```

```

Family: poisson
Links: mu = log
Formula: externalising ~ breast_development_p_sc + adhd_traits_sc + age_sc + ethnrace + inr_sc
Data: analytic_df (Number of observations: 19364)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```
~family_id (Number of levels: 4619)
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.715	0.029	0.656	0.771	1.001	594	1309

```
~obs_id (Number of levels: 19364)
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.369	0.008	0.353	0.385	1.002	1467	2634

```
~participant_id (Number of levels: 5289)
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.725	0.026	0.676	0.777	1.002	629	1094

```
~site_id (Number of levels: 22)
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.112	0.028	0.066	0.175	1.000	3350	2730

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	0.441	0.035	0.371	0.511	1.000	3472
breast_development_p_sc	0.104	0.019	0.067	0.142	1.003	6899
adhd_traits_sc	1.088	0.017	1.055	1.121	1.000	4659
age_sc	-0.129	0.016	-0.160	-0.097	1.002	6171
ethnraceblack_nh	-0.076	0.050	-0.173	0.023	1.000	4312
ethnracehispanic	-0.045	0.047	-0.137	0.048	1.001	5336
ethnraceother	-0.056	0.053	-0.158	0.050	1.000	5217
inr_sc	-0.148	0.024	-0.195	-0.103	1.000	6261
	Tail_ESS					
Intercept		2818				
breast_development_p_sc		3247				

adhd_traits_sc	2980
age_sc	3519
ethnraceblack_nh	3086
ethnracehispanic	3029
ethnraceother	3099
inr_sc	3062

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 5B no interaction effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

A tibble: 2 x 4

Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development (per 1-unit)	6.64	4.22	9.15
2 ADHD traits (per 1-unit)	21.3	20.6	22.0